PROPER FLAT MORPHISMS WITH TRIVIAL DEGREE-ZERO FUNCTION FIELD COHOMOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: both clauses are proved in full, the scheme case being handled through the cited jump-locus and cut-complex lemmas

Editorial note. The label lemma-proper-flat-h0 names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- cohomology and base change for proper flat finite presentation morphisms giving a perfect E = Rf_*O_X whose formation commutes with base change
- the jump-locus lemma: where H^0 of a perfect complex is locally free of rank 1 the locus is universal closed and the map factors
- the diagonal section argument splitting a direct image with a section
- the cut-complex lemma splitting a perfect complex where H^0 fibre-surjects
- the ring fact that an algebra invertible as a module over the base is the base

Source reference. Original reference 0B91: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 6442–6442. Prerequisite pointer invoked at proof line(s) 6852; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BDI: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 6667–6667. Prerequisite pointer invoked at proof line(s) 6859; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BDL: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 6788–6788. Prerequisite pointer invoked at proof line(s) 6860; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0A1U: sources/stacks-chow/tags.txt, source lines 12682–12682. Prerequisite pointer invoked at proof line(s) 6891; it names a standard

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume*

(1) *f is proper, flat, and of finite presentation, and*
(2) *for a morphism $\mathrm{Spec}(k) \to Y$ where k is a field, we have $k = H^0(X_k, \mathcal{O}_{X_k})$.*

Then we have

(a) *$f_*\mathcal{O}_X = \mathcal{O}_Y$ and this holds after any base change,*
(b) *étale locally on Y we have*

$$Rf_*\mathcal{O}_X = \mathcal{O}_Y \oplus P$$

in $D(\mathcal{O}_Y)$ where P is perfect of tor amplitude in $[1, \infty)$.

Proof. It suffices to prove (a) and (b) étale locally on Y, thus we may and do assume Y is an affine scheme. By cohomology and base change (Lemma 0B91) the complex $E = Rf_*\mathcal{O}_X$ is perfect and its formation commutes with arbitrary base change. In particular, for $y \in Y$ we see that $H^0(E \otimes^{\mathbf{L}} \kappa(y)) = H^0(X_y, \mathcal{O}_{X_y}) = \kappa(y)$. Thus $\beta_0(y) \leq 1$ for all $y \in Y$ with notation as in Lemma 0BDI. Apply Lemma 0BDL with $a = 0$ and $r = 1$. We obtain a universal closed subscheme $j : Z \to Y$ with $H^0(Lj^*E)$ invertible characterized by the equivalence of (4)(a), (b), and (c) of the lemma. Since formation of E commutes with base change, we have

$$Lf^*E = R\mathrm{pr}_{1,*}\mathcal{O}_{X \times_Y X}$$

The morphism $\mathrm{pr}_1 : X \times_Y X$ has a section namely the diagonal morphism Δ for X over Y. We obtain maps

$$\mathcal{O}_X \longrightarrow R\mathrm{pr}_{1,*}\mathcal{O}_{X \times_Y X} \longrightarrow \mathcal{O}_X$$

in $D(\mathcal{O}_X)$ whose composition is the identity. Thus $R\mathrm{pr}_{1,*}\mathcal{O}_{X \times_Y X} = \mathcal{O}_X \oplus E'$ in $D(\mathcal{O}_X)$. Thus $\mathcal{O}_X$ is a direct summand of $H^0(Lf^*E)$ and we conclude that $X \to Y$ factors through Z by the equivalence of (4)(c) and (4)(a) of the lemma cited above. Since $\{X \to Y\}$ is an fppf covering, we have $Z = Y$. Thus $f_*\mathcal{O}_X$ is an invertible $\mathcal{O}_Y$-module. We conclude $\mathcal{O}_Y \to f_*\mathcal{O}_X$ is an isomorphism because a ring map $A \to B$ such that B is invertible as an A-module is an isomorphism. Since the assumptions are preserved under base change, we see that (a) is true.

Proof of (b). Above we have seen that for every $y \in Y$ the map $\mathcal{O}_Y \to H^0(E \otimes^{\mathbf{L}} \kappa(y))$ is surjective. Thus we may apply More on Algebra, Lemma 0A1U to see that in an open neighbourhood of y we have a decomposition $Rf_*\mathcal{O}_X = \mathcal{O}_Y \oplus P$ □